

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Unit I Digital Fundamentals

Degree, Semester& Branch: III Semester B.E. ECE B

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic: Introduction to Digital World

Name of the Innovative Practice: Multimedia Approach

Date & Time: 01.07.2019 & 20 Minutes

Description:

Multimedia activities encourage students to work in groups, express their knowledge in multiple ways, solve problems, revise their own work, and construct knowledge. The benefits of multimedia learning is that it takes advantage of the brain's ability to make connections between verbal and visual representations of content, leading to a deeper understanding, which in turn supports the transfer of learning to other situations.

Learning Outcomes:

1. The students will be able to understand the digital fundamentals in the present contemporary world.
2. The students can be easily identified the impact and importance of different techniques used in the digital world

Justification for choosing the topic using Multimedia approach activity:

Multimedia is one of the most engaging mediums of the modern day. A multimedia approach is a quick way of gathering information on how well our students are meeting their learning objectives. Multimedia is like simple intervention and it can make a significant improvement in the knowledge of engineering students and help them to develop interest in Combinational Logic Circuits. It enables students to apply Imagination & think individually about a topic or answer to a question.

Implementation (Plan & Execution) with Proof:

The suitable multimedia for Digital World was displayed to the students and explained about the different digital devices used in our daily life and their applications.

Innovative Teaching Method Execution

Introduction to Digital World - Multimedia Approach (Khan Academy)



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	1					2	2		

References:

- <https://www.khanacademy.org/computing/computer-science/how-computers-work2/v/khan-academy-and-codeorg-circuits-logic>
- M. Morris Mano and Michael D. Ciletti, "Digital Design", 5th Edition, Pearson, 2014.

CO1: The students will be able to apply the digital fundamentals in the present contemporary world.

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Innovative Practices Description

Unit II Combinational Circuit Design

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic: Combinational Circuits

Name of the Innovative Practice: Quiz

Date & Duration: 07.08.2019

ICT Tool Used: Online Google Forms

Description:

A quiz can be defined as a game or brain teaser to test knowledge. It can contain an element of competition when participants play against each other to get the highest score, which makes helps participants become more engaged. The quiz consists of a large variety of Question types, including multiple choice, true-false, short answer and drag and drop images and text. These questions are kept in the Question bank and can be re-used in different quizzes.

Learning Outcome:

- The students will be able to understand the different functionalities of Combinational Logic Circuits.

Justification for choosing the topic using Quiz activity:

The present study attempts to introduce quiz as an innovative learning method. A quiz is a quick way of gathering information on how well our students are meeting their learning objectives. Quiz is like simple intervention and it can make a significant improvement in the knowledge of engineering students and help them to develop interest in Combinational Logic Circuits.

Implementation(Plan & Execution) with Proof:

Quiz is one of the assessment tool with the following characteristics:

1. Specific questions are asked, specific answers are required; either in a word or in a couple of sentences.
2. The correct answers are pre-determined, not affected by students opinionating or reasoning.
3. The exercise is conducted in a controlled, monitored framework (a classroom or an online quiz area), usually within a short time limit.

In my subject quiz was conducted for Unit-II Combinational Logic Circuits at the completion of unit through Google forms. 20 questions are posted to the students College email id using Google forms. The students are answered the Questions within a week. The screen shot of the students marks, questions and answers are shown below.

Innovative Teaching Method Execution

Combinational Circuits - Quiz

U-II Combinational Circuit Design (Responses) ☆					
File Edit View Insert Format Data Tools Form Add-ons Help All changes saved in Drive					
A. 10111100					
A	B	C	D	E	F
Score	Name	Register Number	1. A combinational logic c	2. In a two-gate half adde	3. The Boolean SOP expr
6 / 20	Sundhar	953618106064	B. False	C. XOR	D. None of these
19 / 20	RAMKISHORE K.R	19LE106302	B. False	A. AND	C. A'B'C+ABC'
17 / 20	M Sathis Kannan	953618106038	B. False	A. AND	C. A'B'C+ABC'
16 / 20	N shambhireddy	953618106044	B. False	A. AND	C. A'B'C+ABC'
18 / 20	R.santhosh	953618106055	B. False	A. AND	C. A'B'C+ABC'
19 / 20	Sai Siddarth R	953618106053	B. False	A. AND	C. A'B'C+ABC'
16 / 20	praveenkumar.p	953618106047	B. False	A. AND	C. A'B'C+ABC'
17 / 20	RAMASURESH M	953618106051	B. False	A. AND	C. A'B'C+ABC'
8 / 20	PAVITHRA	953618106045	B. False	C. XOR	B. AB'C+ABC'
13 / 20	NARMADHADEVI	953618106039	B. False	A. AND	C. A'B'C+ABC'
20 / 20	nisha	953618106042	B. False	A. AND	C. A'B'C+ABC'
8 / 20	Nithish Balaji	953618106043	B. False	B. OR	C. A'B'C+ABC'
4 / 20	Shanmugam	953618106059	B. False	D. NAND	B. AB'C+ABC'

Reflective Critique:

- Benefits:**

The quiz was conducted with multiple choices. The answers reflect the students' understanding, ability to recall, as well as the ability to apply a concept learnt in the class. Fairness is implemented in the objective nature of the quiz, i.e students do not feel that they are being judged by a different measure than that of any other student. Students can display their knowledge by overcoming certain difficulties faced by them by a paper assignment for eg. Language skills, paper anxiety, viva anxiety, word limitations etc. After the quiz, students gain knowledge by learning the correct answers to any questions they missed, can highlight the areas requiring improvement or reconsider their study habits.

- Challenges:**

Few students take more than the time given for this activity

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	1	3	3	3			1		1		

Reference:

- M. Morris Mano and Michael D. Ciletti, "Digital Design", 5th Edition, Pearson, 2014.

CO2: The students will be able to design and implement various combinational digital circuits using logic gates.

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Unit III Synchronous Sequential Circuits

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic: Sequential Logic Circuits

Name of the Innovative Practice: Mind Map

Date & Duration: 30.08.2019 & 15 Minutes

ICT Tool Used: Black Board

Description:

A mind map is a diagram used to visually organize information. A mind map is hierarchical and shows relationships among pieces of the whole. Mind mapping in a learning environment which requires an enhanced way of thinking, learning and presenting. Mind Mapping enhances the learning capacity in terms of number of ideas generated as well as improves presentation focus. Mind maps are used to generate, visualize, structure and classify ideas. Mind maps go under variety of names. They are also known as concept maps, semantic mapping, knowledge mapping, think links, graphic organizers or cognitive maps.

Learning Outcome:

- The students will be able to visualize the concepts and organize information.

Justification for choosing the topic using Mind Map activity:

Mind mapping is a creative way to set goals, solve problems and design action plans. It quickly records ideas in a free-form way. When groups use mind mapping, the thoughts of each participant easily trigger ideas in others. This dynamic group interaction encourages breaking free of old patterns to uncover new and innovative approaches.

Implementation(Plan & Execution) with Proof:

Mind mapping activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a mind map, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

In my subject Mind map was conducted for Unit-III Synchronous Sequential Circuits at the completion of unit. The students are voluntarily drawn the map. The sample copy of the mind map was done by the students using green board was shown below.

Innovative Teaching Method Execution

Synchronous Sequential Circuits – Mind Map



Reflective Critique:

- **Benefits:**

The benefits of Mind Maps are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information using key words, and then make connections between facts and ideas visually – keeping all of your topic thoughts together on one sheet. It made key note making easier to students, as it reduces pages of notes into one single side of paper. Also mind map made slow learners to remember the information more quickly.

- **Challenges:**

Normally teachers will give longer explanations in the notes section of the topic. At the completion of the unit, teacher will revise the unit. The students are called randomly to draw the map to indicate relationships between the topics in Sequential logic circuits. I planned the activity for 10 minutes only. But in real scenario it takes 15 minutes to complete the whole topics in Unit-III.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	3	2	3	1				1	3	1		

Reference:

- M. Morris Mano and Michael D. Ciletti, “Digital Design”, 5th Edition, Pearson, 2014.

CO3: The students will be able to design and analyze synchronous sequential circuits

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
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Innovative Practices Description

Unit IV ASynchronous Sequential Circuits

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic : Introduction about Asynchronous sequential circuits

Name of the Innovative Practice: Think-Pair-Share

Date & Duration: 04.09.2019 & 20 Minutes

ICT Tool Used: -

Description:

- T (Think): Teacher asks a specific question about the topic. Students "think" about what they know or have learned, and come up with their own individual answer to the question. [Takes 2 Minutes].
- P (Pair): Teacher asks another question, related to the previous one, that is suitable to deepen the students' understanding of the topic. Each student is paired with another student. They share their thinking with each other and proceed with the task. [Takes 3 Minutes].
- S (Share): Students share their thinking (or solution) with the entire class. Teacher moderates the discussion and highlights important points. [Takes 10 minutes].

Learning Outcome:

- The students were able to explain conceptual understanding and analyse the pros and cons of different ways of sequential circuits

Justification for choosing the topic using Think-Pair-Share activity:

- It helps students to think individually about a topic or answer to a question.
- It teaches students to share ideas with classmates and builds oral communication skills.
- It helps focus attention and engage students in comprehending the reading material.

Implementation (Plan & Execution) with Proof:

Think-Pair-Share innovative practice conducted for II year ECE B students, after explained the concept of Synchronous sequential circuits with an example of Counters, Shift registers. First, I asked the students to think about counter circuit based on common clock pulse for 2 minutes. Then I make them as a pair to discuss their neighbour's and asked the students to discuss about asynchronous sequential circuits for 3 minutes. Finally I asked the one of the pair to explain the concept to whole class for further discussion. The students from group share their points and participated in the discussion for 10 minutes.

Innovative Teaching Method Execution

Introduction about Asynchronous sequential circuits - Think-Pair-Share



Reflective Critique:

- **Benefits:**

Think-pair-sharing forces all students to attempt an initial response to the question, which they can then clarify and expand as they collaborate. It also gives them a chance to validate their ideas in a small group before mentioning them to the large group, which may help shy students feel more confident participating.

- **Challenges:**

Four to six students are not shown interest to actively participate in groups. I planned the activity for 15 minutes. But in Class room it takes 20 minutes.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	3	2	2	2				1	2	2		

Reference:

- M. Morris Mano and Michael D. Ciletti, "Digital Design", 5th Edition, Pearson, 2014.

CO4: The students will be able to design and implement asynchronous sequential circuits

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Innovative Practices Description

Unit V Memory Devices and Digital Integrated Circuits

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic : RAM & ROM

Name of the Innovative Practice: Animation Video

Date & Duration: 16.09.2019 & 20 Minutes

ICT Tool Used: LCD Projector

Description:

Use of animation in special education has gained great attention recently. Student engagement and understanding of materials is given more emphasis in today's education over spoon feeding the facts. Therefore, using black-boards or the typical lecture methods are not adequate to teach science and other related subjects. The evolution of a process can be conveyed better with animation videos. It can also be videos of demonstration of an idea or an application side of a theory or an interview with a scientist, tutorial by a subject expert and more.

Learning Outcome:

- The students will be able to easily identify the different types of RAM used for store the data in the memory device.
- The students will be able to explain the application of RAM and ROM.

Justification for choosing the topic using Video Clip activity:

Video is one of the most engaging mediums of the modern day. Animation videos are one of the digital tools for classroom. It enables students to apply Imagination & think individually about a topic or answer to a question. It helps to promote creativity and motivate the students to do mini projects using memory devices

Implementation (Plan & Execution) with Proof:

The suitable animated video for RAM and ROM was displayed to the students and explained about the different types of memory devices and their applications.

Innovative Teaching Method Execution

RAM & ROM - Video Clip



Reflective Critique:

- **Benefits:**

- The students are more interested to watch animated video, apart from theory concept.
- Using this activity strengthen the students talent and skill set.
- It helps the students to communicate stories, ideas and concepts in a creative and original way.

- **Challenges:**

It takes more time to choose relevant animated videos for easily understanding the concepts of different types of memory devices.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	3	3	2	1				1		1		

Reference:

- <https://www.youtube.com/watch?v=UaFSsD0LPS8>
- M. Morris Mano and Michael D. Ciletti, "Digital Design", 5th Edition, Pearson, 2014.

CO5: The students will be able to apply the concepts of memory devices and programmable logic devices in Integrated Circuits